Automation in Construction 185 (2026) 106840



Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon





Automated generation of building energy model geometry from BIM-derived space and opening volumes

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ARTICLE INFO

Keywords:
Building information model
Industry foundation classes
Building energy model
Building geometry
Second-level space boundaries

ABSTRACT

Accurate geometry is critical in Building Energy Models (BEM), particularly the second-level space-boundary (2LSB) surfaces that define thermal zone boundaries and support energy-transfer calculations. Existing 2LSB generation methods require detailed design input and are vulnerable to modeling errors and incomplete geometry, creating a persistent gap in automating the production of simulation-ready BEM geometry from Building Information Models (BIM). This paper addresses this gap by presenting an automated method for generating 2LSB surfaces directly from the space- and opening-volume data available in IFC files. The method requires only a user-defined construction-thickness threshold and intersection-free space volumes, while relying on the input space and opening data to be pre-processed and free of geometric inconsistencies. Its design enables scalable and robust handling of complex building geometries once properly prepared. Two case studies, a university building and an office building, demonstrate the approach's scalability and effectiveness, including its accurate treatment of opening volumes.

1. Introduction

Accurate building energy simulations depend on reliable Building Energy Models (BEMs[1]), in particular on their geometric representation of thermal zone boundaries, commonly referred to as the second-level space boundary (2LSB) surface set [1]. Conventional methods for generating this geometry rely heavily on detailed construction elements, making them labor-intensive and error-prone, especially for large or complex buildings. To address this limitation, an automated method that derives BEM geometry directly from space and opening volume definitions available in BIM[2] IFC files, is proposed. By operating on space- and opening-based representations rather than detailed architectural components, the method enables efficient, robust, and scalable generation of simulation-ready BEMs with minimal input requirements, making it suitable for federated and evolving BIM workflows.

Meaningful energy simulations require BEMs to combine multiple data sources, including weather, material properties [2], internal loads, schedules, and energy systems [3], together with an accurate geometric representation of building thermal exchange surfaces [4]. As illustrated in Fig. 1, these surfaces correspond to the type 2a and 2b second-level space boundaries extracted from an architectural layout. Together, they form the 2LSB surface set, which provides the geometric basis for thermal exchange calculations.

Simulation engines such as EnergyPlus model building constructions as zero-thickness surfaces, placing external boundaries at the building envelope and internal boundaries at construction centerlines to represent inter-zone heat transfer [5]. These surfaces can be directly derived from the 2LSB surface set, which provides the required geometric attributes, including orientation, area, adjacency, and tilt. Despite differing conventions for boundary placement, the 2LSB representation constitutes a minimal and complete geometric abstraction that supports consistent translation to simulation platforms such as EnergyPlus, TRNSYS [6], and Modelica [7], thereby acting as a unified interface between BIM data and energy simulation models.

The 2LSB surface set can be generated manually from architectural plans, a time-consuming task, or automatically using BIM data. When available, BIMs allow extraction of relevant information in a process known as BIM2BEM or BIM-based BEM [8]. With the growing

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[1] In this paper, BEM refers to both Building Energy Modeling and Building Energy Model.
[2] In this paper, BIM refers to both Building Information Modeling and Building Information Model.

https://doi.org/10.1016/j.autcon.2026.106840
Received 29 September 2025; Received in revised form 7 February 2026; Accepted 12 February 2026
Available online 5 March 2026
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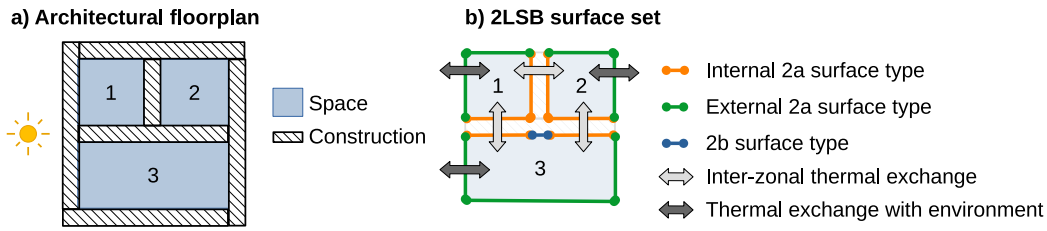


Fig. 1. 2LSB surface set and its respective architectural floor plan.

Nomenclature

Symbol	Definition
$\mathcal{A}$	Polygon set $\mathcal{A}$
$\mathcal{A} \leftarrow \mathcal{B}$	Polygon set $\mathcal{A}$ is replaced by polygon set $\mathcal{B}$
A	Polygon A (Coplanar ordered 3D point sets of outside and hole contour(s))
$\hat{n}_A$	Unit normal vector of polygon A using the right-hand rule
$-A$	Inverse of polygon A, with $\hat{n}_{-A} = -\hat{n}_A$ (Coplanar 3D point sets of polygon A, with their point order reversed)
$\langle , \rangle$	Inner product (dot product)
$\setminus_p$	Subtraction of coplanar polygons
$\cap_p$	Intersection of coplanar polygons
$\cup$	Union of polygon sets
$\lvert \ \rvert$	Absolute value
$\emptyset$	Empty set
[...] = F(...)	Function F and its output [ ] and input ( ) sets

adoption of BIM in design, such automated approaches can streamline the preparation of simulation-ready models for building energy analysis [9].

A complete BIM2BEM workflow encompasses diverse data transformations from geometric data transformations involved in the generation of the 2LSB surface set, to active system data transformations related to HVAC systems [10]. In practice, these workflows encounter BIM errors, due to design mistakes [11] or export flaws [12]. Current BIM2BEM efforts focus on improving interoperability [13,14] and automation [15,16], including IFC-based frameworks [17], yet key issues remain: inaccuracies, missing data, and incomplete BIM content [18]. As a result, geometry, material, thermal zoning [7], and HVAC information often require manual correction, and no fully robust, universal BIM2BEM workflow exists [19].

A major challenge in BIM2BEM workflows is the automatic derivation of a consistent 2LSB surface set from BIM geometry. Although commercial BIM tools can export first- or second-level space boundaries, these outputs are often incomplete or inconsistent for energy simulation, exhibiting issues such as misaligned or duplicated surfaces, inverted normals, or missing space adjacency information. Since such exports typically depend on detailed construction geometry, any inaccuracies in walls, slabs, or roofs propagate directly into the resulting 2LSB surface set, requiring costly post-processing and manual correction. To address these limitations, several algorithmic approaches have been proposed. Graph-based methods represent spaces and constructions as nodes with adjacency relations encoded as edges, enabling structured surface extraction [20–22]. Projection-based approaches instead derive boundaries through geometric intersections or projections of space and construction elements, as in CBIP and related methods [23–25]. More robust geometry-driven approaches, such as Fichter's algorithm [26], rely exclusively on construction geometry and use projection, clipping, and ray-tracing to classify interior and exterior surfaces, making them more tolerant to common BIM geometric defects. A comprehensive comparison of these methods, including their assumptions and limitations, is provided in [26,27].

Existing methods largely depend on explicit geometric definitions of construction elements (e.g., walls, slabs, roofs) to derive 2LSB surfaces. In practice, such definitions often contain inconsistencies or missing information [27], which complicate the BIM2BEM automation and require extensive data quality control and manual correction to obtain simulation-ready BEMs [28]. Increasing input data requirements further amplify this burden. In addition, construction-based approaches struggle to represent virtual internal partitions introduced for zoning or simulation purposes, such as separations within open-plan spaces. Our method supports the automatic generation of corresponding virtual 2LSB-2a surfaces at user-defined virtual partition locations [29], without requiring physical wall elements. When explicit space volumes are absent, enclosing spaces can also be inferred from surrounding construction geometry [30], ensuring broader applicability. Overall, this work advances a construction-independent, space-centric approach for generating 2LSB surface sets directly from BIM-derived data.

The remainder of the paper is structured as follows: Section 2 introduces the mathematical foundations of the approach, including two core functions that underpin the methodology. Section 3 details the algorithmic process used to generate the 2LSB surfaces, outlining key principles and steps. Section 4 analyzes the computational complexity and accuracy of the method. Section 5 describes on the implementation of the proposed workflow within a software framework, highlighting data processing strategies and computational considerations. Section 6 describes the geometric data prerequisites of the method, and Section 7 presents case studies on real-world buildings to evaluate the method's robustness and performance in complex scenarios. Sections 8–10 resume the work with a discussion of results, current limitations, and directions for future development toward broader integration within BIM2BEM workflows.

2. Mathematical background

Initially, a set of functions is introduced in the following subsections, that are used in the main methodology described in Section 3.

2.1. Intersection projection functions

In this section, fundamental functions used to perform basic geometric operations, are introduced. These have been discussed in more detail in our previous work on regular, higher-order, and district-level 2LSB surface calculation [23,31,32], but are included here to support the new algorithm and ensure the paper is self-contained. Specifically, they include the Intersection Projection function (IP_d) which applies intersection and projection operations to 3D polygon pairs, and the intersection projection on polygon sets function (IPS_d), which applies IP_d to sets of polygons. The subscript d is used in IP_d and IPS_d function calls, as both functions use a thickness threshold parameter d.

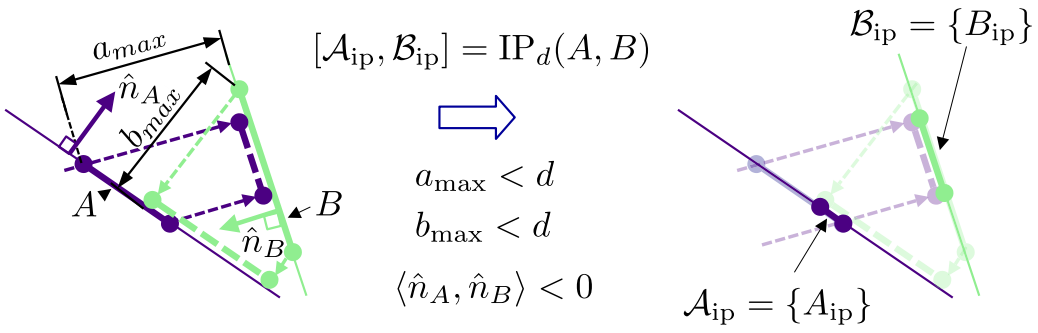


Fig. 2. Intersection projection function on two polygons A and B within the threshold distance d.

2.1.1. *Intersection projection function* (IP_d)

The intersection projection function (IP_d), acts on a polygon pair (A, B) that has opposite oriented polygons ($\langle \hat{n}_A, \hat{n}_B \rangle < 0$) and returns two intersection polygon sets $\mathcal{A}_{\text{ip}}$ and $\mathcal{B}_{\text{ip}}$:

$$[\mathcal{A}_{\text{ip}}, \mathcal{B}_{\text{ip}}] = \text{IP}_d(A, B). \tag{1}$$

IP_d function performs its operation in two steps:

1. **Projection step:** Polygon A is orthogonally projected onto the plane of polygon B, forming a new polygon A_{p} using the unit normal of polygon B $\hat{n}_A$ for the projection, as indicated by the dashed arrows of Fig. 2. Similarly, polygon B is projected onto the plane of polygon A using the unit normal vector $\hat{n}_A$, resulting in a new polygon B_{p}.
2. **Intersection step:** The projected polygon A_{p} is intersected with the original coplanar polygon B, producing the output polygon set $\mathcal{A}_{\text{ip}}$. Likewise, the projected polygon B_{p} is intersected with the original coplanar polygon A, yielding the output polygon set $\mathcal{B}_{\text{ip}}$.

The function IP_d operates using a user-defined parameter d, which serves as a distance threshold. This threshold defines the maximum allowable distance from any point of one polygon to the plane of the other, in both directions. If a point on either polygon lies beyond this threshold relative to the other's plane, IP_d returns $\emptyset$. Fig. 2 demonstrates how IP_d behaves when applied to two oppositely oriented polygons, A and B, that are within the threshold distance d. As shown, the maximum distance from any point on A to the plane of B ($a_{\max}$), and from any point on B to the plane of A ($b_{\max}$), are both less than the threshold d.

Finally, IP_d preserves the orientation of the original polygons in the respective output polygon sets. In other words: $\forall A_{\text{ip}} \in \mathcal{A}_{\text{ip}} \quad \hat{n}_{A_{\text{ip}}} = \hat{n}_A$ and $\forall B_{\text{ip}} \in \mathcal{B}_{\text{ip}} \quad \hat{n}_{B_{\text{ip}}} = \hat{n}_B$.

2.1.2. *Intersection projection on polygon sets function* (IPS_d)

The intersection projection on polygon sets function (IPS_{d}) applies the previous IP_d function on all possible polygon pairs (A, B) taken from two input polygon sets $\mathcal{A}$ and $\mathcal{B}$ and returns the intersection projection polygon set $\mathcal{AB}_{ip}$, as described in the following algorithm:

Algorithm 1 *Intersection projection on polygon sets function:* $\mathcal{AB}_{\text{ip}} = \text{IPS}_d(\mathcal{A}, \mathcal{B})$

```
Initialize the result set: AB_ip ← ∅
for all A ∈ 𝒜, B ∈ ℬ do
    [A_ip, B_ip] = IP_d(A, B)
    if A_ip ≠ ∅ and B_ip ≠ ∅ then
        AB_ip ← AB_ip ∪ A_ip ∪ B_ip
    end if
end for
```

When the input sets coincide ($\mathcal{A} = \mathcal{B}$), polygon pairs referring to the same polygon ($A = B$) are excluded from the intersection–projection operation, and commutative pairs are evaluated only once. The operator IPS_d applies the intersection–projection function IP_d to all polygon pairs (A, B) with $A \in \mathcal{A}$ and $B \in \mathcal{B}$ whose outward unit normal vectors satisfy $\langle \hat{n}_A, \hat{n}_B \rangle < 0$ and the max distances $a_{\max}$ and $b_{\max}$ are smaller than the distance threshold d ($a_{\max} < d$ and $b_{\max} < d$). By definition of the input, the polygon sets $\mathcal{A}$ and $\mathcal{B}$ correspond to distinct, non-overlapping space or opening volumes; therefore, the case $\mathcal{A} \neq \mathcal{B}$ with a non-empty geometric intersection between the two sets cannot occur. The operator is thus applied only to identify polygon pairs that are within a distance smaller than the threshold d and oriented toward each other, as required for the detection of 2LSB-2a surface pairs in Algorithms 3 and 5.

2.2. *Polygon set subtraction function* (PSS)

The polygon set subtraction function acts on two input polygon sets $\mathcal{A}$ and $\mathcal{B}$ and returns a subtraction polygon set $\mathcal{S}$ that contains the remaining polygons after subtracting from polygons $A \in \mathcal{A}$ coplanar polygons $B \in \mathcal{B}$. The coplanarity between polygons A and B is assessed by verifying that their unit normal vectors satisfy ($|\langle \hat{n}_A, \hat{n}_B \rangle| = 1$), and by confirming that the distances of their supporting planes from the origin are equal ($d_A = d_B$). Algorithm 2 describes the polygon set subtraction function and is illustrated in Fig. 3. In short, for each polygon $A \in \mathcal{A}$, a corresponding set of remaining polygons $\mathcal{R}_A$ is initialized as $\mathcal{R}_A \leftarrow \{A\}$. Then, for every polygon $B \in \mathcal{B}$, the set $\mathcal{R}_A$ is updated by replacing each polygon $R \in \mathcal{R}_A$ with the result of subtracting any coplanar polygon B from it. Finally, the output set $\mathcal{S}$, which is the subtraction polygon set, is formed as a union of all remaining sets $\mathcal{R}_A$, $\forall A \in \mathcal{A}$. This surface set does not necessarily form a closed shell.

Algorithm 2 *Polygon Set Subtraction function:* $\mathcal{S} = \text{PSS}(\mathcal{A}, \mathcal{B})$

```
S ← ∅
for all A ∈ 𝒜 do
    R_A ← {A}
    for all B ∈ ℬ such that |⟨n̂_A, n̂_B⟩| = 1 ∧ d_A = d_B do
        R_A ← ⋃_{R∈R_A} {R \_p B}
    end for
    S ← S ∪ R_A
end for
```

3. Proposed methodology

The proposed methodology consists of four sequential stages, taking one or more BIMs in IFC4 format [33] as input and producing the

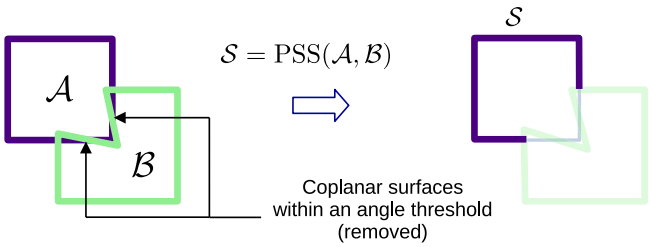


Fig. 3. Polygon set subtraction function operation on two polygon sets $\mathcal{A}$ and $\mathcal{B}$.

computed 2LSB surface set data as output. In particular, the method operates on spaces and openings as semantically defined in IFC classes, namely IfcSpace and IfcOpeningElement. While the approach could in principle be extended to other model types, this would require model-specific preprocessing to derive equivalent semantic and geometric entities, which is outside the scope of this work. The four stages of the methodology are illustrated in Fig. 4.

- **A. Input:** In this stage, the input is one or more (in the cases of federated models) IFC4.3 [33] models containing geometric descriptions of building space and opening volumes. These descriptions can be either procedural or explicit.
- **B. Data Preprocessing:** The input IFC files pass through three preprocessing steps: (B1) Parsing, (B2) Filtering, and (B3) B-rep generation that are illustrated in block B in Fig. 4 and described in detail in Section 3.2.
- **C. Main processing:** The boundary representations extracted in Stage B are processed in four steps. The first three steps (displayed as C1, C2 and C3 parts of block C in Fig. 4), focus on generating 2LSB surfaces of types 2a and 2b, which represent non-opening elements. In the fourth and final step (displayed as C4 part of block C in Fig. 4), the algorithm generates 2LSB-2a surfaces corresponding to openings. At the same time at this step, the previously generated 2LSB-2a surfaces for non-opening elements are updated by subtracting the newly created opening surfaces. The main processing stage is described in detail in Section 3.3.
- **D. Output:** In the final stage, the detected 2LSB 2a surfaces are aggregated to form the complete 2LSB surface set, which is then exported in three formats:

    1. **XML**, as a native default output, structured according to a predefined XSD schema.
    2. **OBJ**, for visual validation.
    3. **gbXML**, to ensure compatibility with external building energy simulation tools.

    Although the current implementation does not document IFC enrichment with the generated 2LSB surface set, [34] reports that such an enrichment process has already been developed, using the XML output as its input.

The following subsections provide detailed explanations of each processing stage, along with an analysis of their algorithmic processes where applicable.

3.1. Input stage

In large-scale real-world building projects, models are typically delivered in a federated setup, where different building parts are exported as separate IFC files. To ensure compatibility with the proposed methodology, these files must be spatially coordinated and aligned using a common project base point and reference coordinate system. Each federated model is expected to contain spaces and openings as semantically defined in IFC (IfcSpace and IfcOpeningElement classes). During the input stage, one or more such coordinated IFC files are aggregated to produce a unified set of building input data.

3.2. Data preprocessing stage

Because the IFC files collected during the input stage may contain both procedural and explicit geometry representations, their geometric content must first be extracted and normalized before further processing. This stage operates on spaces and openings as semantically defined in IFC, namely IfcSpace and IfcOpeningElement, and prepares their geometry for the subsequent computation steps.

The preprocessing workflow consists of three conceptual steps (Block B in Fig. 4). First, the geometric descriptions of all IFC elements are deserialized to make their underlying representations accessible (Part B1). Second, the geometric data corresponding specifically to space and opening entities are isolated from the full set of architectural elements contained in the model (Part B2).

Finally, all extracted geometries are transformed into boundary representations (B-reps), which are required by the core methodology (Part C). Any procedural definitions (e.g., swept solids, extrusions, or CSG-based representations) are converted into boundary surfaces sets, whereas explicit geometries are passed through directly. The result of this step is a collection of polygonal boundary representations with consistently oriented outward-facing normals.

At the end of this stage, the geometry is represented as polygon sets defining closed solids. Space volumes are denoted by N polygon sets $\mathcal{SP}_i$, for $i = 1, \dots, N$, and opening volumes by M polygon sets $\mathcal{OP}_j$, for $j = 1, \dots, M$. Together with the prescribed thickness threshold d, these sets form the input to the subsequent processing stage.

3.3. Main stage

The main stage transforms the boundary-representation (B-rep) geometry of spaces $(\mathcal{SP}_1, \dots, \mathcal{SP}_N)$ and openings $(\mathcal{OP}_1, \dots, \mathcal{OP}_M)$ into the three surface sets required for a Building Energy Model (BEM):

$\mathcal{I}_{2a}$– internal (inter-zone) 2LSB-2a surfaces
$\mathcal{E}_{2a}$– external (exposed) 2LSB-2a surfaces
$\mathcal{O}_{2a}$– opening-related 2LSB-2a surfaces

All operations share a user-defined thickness threshold d that distinguishes neighboring surfaces from distinct constructions. This can be expressed mathematically by the following main function F_m:

$$[\mathcal{I}_{2a}, \mathcal{E}_{2a}, \mathcal{O}_{2a}] = \mathrm{F_m}(\{\mathcal{SP}_1, \dots, \mathcal{SP}_N\}, \{\mathcal{OP}_1, \dots, \mathcal{OP}_M\}, d) \quad (2)$$

The three output 2LSB surface polygon sets of $\mathrm{F_m}$ ($\mathcal{I}_{2a}$, $\mathcal{E}_{2a}$ and $\mathcal{O}_{2a}$), are obtained using four processing steps (displayed as parts C1, C2, C3, and C4 in block C in Fig. 4), which use two fundamental functions analyzed in Section 2. These processing steps are described in the following subsections.

3.3.1. Iternal 2LSB-2a surface generation

The internal second-level space boundaries of type 2A are extracted from the boundary representations of the space volumes using the polygon set intersection projection function (IPS_d), described in Section 2.1.2. In short, IPS_d projects the boundary surfaces of one space volume to the boundary surfaces of another and intersects the projections with the original surfaces. IPS_d is applied to all possible pairs of space volumes ($\mathcal{SP}_i$, $\mathcal{SP}_j$) according to the following Algorithm 3:

The set $\mathbb{I}_N$ is the set $\{1, \dots, N\}$ where N is the number of total space volumes defined in the BIM. The condition $i < j$ ensures that commutative space pairs (i, j) and (j, i) are considered once. The thickness threshold value d is also transferred in the function $\mathrm{I}_{2a,d}$. The internal

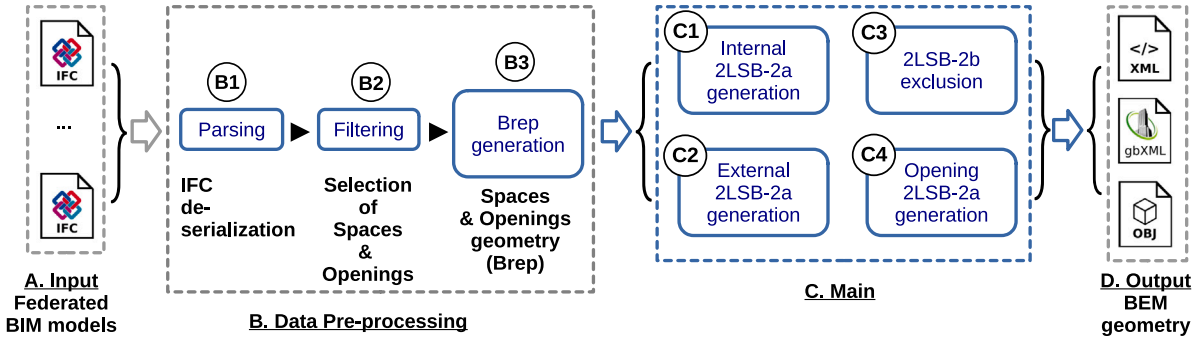


Fig. 4. Block diagram of the proposed workflow.

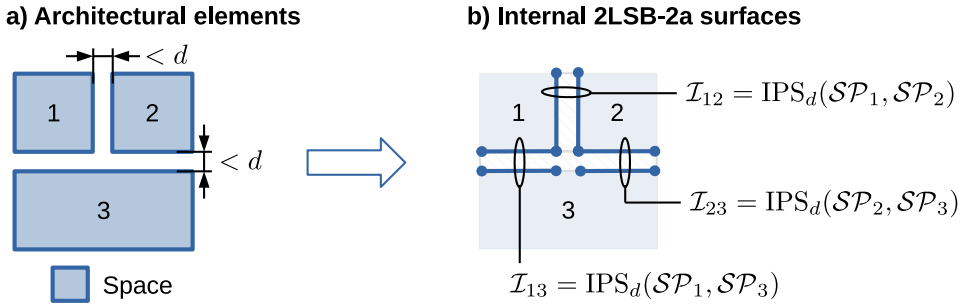


Fig. 5. Internal 2LSB surface generation example from three space volumes, using the threshold distance d.

Algorithm 3 *Internal 2LSB-2a extraction* $\mathcal{I}_{2a} = \mathrm{I}_{2a,d}(\{\mathcal{SP}_1, ..., \mathcal{SP}_N\})$

$\mathcal{I}_{2a} \leftarrow \{\emptyset\}$
for all $(i, j) \in \mathbb{I}_N \times \mathbb{I}_N \mid i \neq j, i < j$ **do**
  $\mathcal{I}_{ij} = \mathrm{IPS}_d(\mathcal{SP}_i, \mathcal{SP}_j)$
  if $\mathcal{I}_{ij} \neq \emptyset$ **then**
    $\mathcal{I}_{2a} \leftarrow \mathcal{I}_{2a} \cup \mathcal{I}_{ij}$
  end if
end for

2LSB surface extraction is illustrated in Fig. 5. In this illustration, the boundary representations of three space volumes $(\mathcal{SP}_1, \mathcal{SP}_2, \mathcal{SP}_3)$ are used as input to $\mathrm{I}_{2a,d}$ a function, that produces three internal 2LSB surface pairs $(\mathcal{I}_{12}, \mathcal{I}_{23}, \mathcal{I}_{13})$, corresponding to all possible space pairs.

3.3.2. External 2LSB-2a surface generation

The external 2LSB surfaces of type 2a $(\mathcal{E}_{2a})$ are generated using the PSS function, as defined in Section 2.2 and illustrated in Fig. 6. Specifically, for each space volume i, the external 2LSB-2a surfaces $(\mathcal{E}_i \subset \mathcal{E}_{2a})$, are obtained by applying the PSS function to subtract the internal 2LSB-2a surfaces $(\mathcal{I}_i \subset \mathcal{I}_{2a})$, obtained in the previous step for space i, from the B-rep of the space $(\mathcal{SP}_i)$. This subtraction can be expressed mathematically as:

$$\mathcal{E}_i = \mathrm{PSS}(\mathcal{SP}_i, \mathcal{I}_i) \tag{3}$$

The set of polygons that contains the internal 2LSB-2a surfaces referring to space i can be obtained by the following union of sets:

$$\mathcal{I}_i = \left(\bigcup_{\forall k<i} \mathcal{I}_{ki}\right) \cup \left(\bigcup_{\forall m>i} \mathcal{I}_{im}\right) \tag{4}$$

As shown in Fig. 6, for space 3, the PSS function generates an internal 2LSB-2b surface in the set $\mathcal{E}_3$, which should not be part of the external 2LSB-2a surfaces for this space. Surfaces like these are eliminated from the external 2LSB-2a sets through a 2LSB-2b exclusion process, which is detailed in the next subsection.

3.3.3. 2LSB-2b surface exclusion

This section describes the detection of 2LSB-2b surfaces, which are excluded from the external 2LSB-2a surface set generated in the previous processing step. The detection and exclusion process relies on computing the alpha wrap [35] of the building's space volumes. The alpha wrap is a triangulated boundary representation that encapsulates all space solids while avoiding entering the small gaps between them, where the internal walls are located. This avoidance is ensured by setting the alpha parameter of the wrap to be greater than the maximum allowable thickness threshold d, as defined in Sections 2.1.2 and 3.3.1. The alpha wrap is illustrated for three space volumes in Fig. 7. In this figure, the central 2LSB-2b surface is not attached to the alpha wrap surfaces and therefore is excluded.

The 2LSB-2b exclusion process is described by Algorithm 4. Each polygon E in the external 2LSB-2a set $(\mathcal{E}_{2a})$ is tested for coplanar intersection $(\cap_p)$ with each triangle of the alpha wrap $W \in \mathcal{W}_d$. The alpha wrap is controlled by two parameters, $\alpha > 0$ and $\mathrm{offset} > 0$, whose selection depends on the geometric characteristics of the building envelope, which typically consists of predominantly planar surfaces.

The parameter α regulates the level of geometric detail captured by the wrap: larger values yield smoother approximations with fewer triangles, while smaller values increase geometric fidelity and mesh complexity. The offset parameter defines a minimum separation between the original geometry and the wrap and must be strictly positive. Smaller offset values (corresponding to larger reciprocal values in the CGAL implementation) produce tighter wraps but increase sensitivity to numerical noise and the number of generated triangles. In practice, both parameters are chosen jointly to ensure that the wrap fully encloses the geometry, remains sufficiently close to enable reliable coplanar intersection tests, and maintains manageable computational complexity. For typical building models with largely planar envelopes, moderate values of α combined with small, non-zero offsets were found to provide stable results.

If an intersection is detected, E is added to the updated external 2LSB-2a set $\mathcal{E}'_{2a}$; otherwise, it is assigned to the 2LSB-2b set $\mathcal{I}_{2b}$. In three dimensions, a coplanar intersection yields a non-empty polygon set

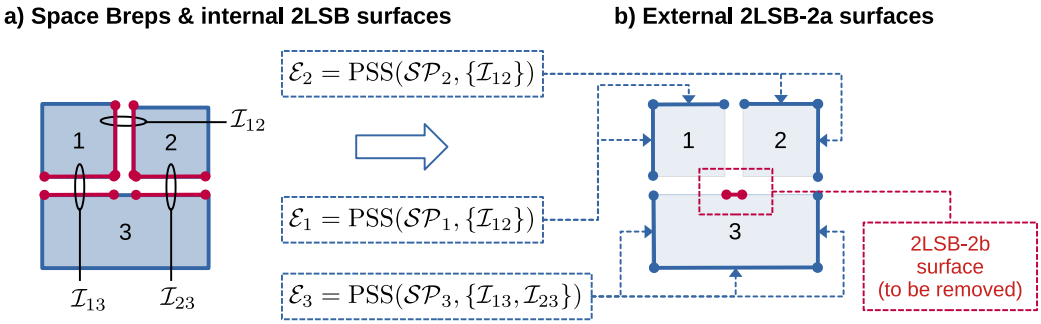


Fig. 6. External 2LSB surface generation example from three space volumes, using the PSS function.

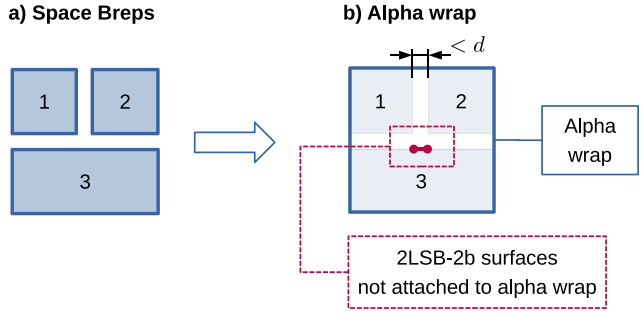


Fig. 7. 2LSB-2b exclusion example, using the alpha wrap.

when the polygons share a common supporting plane and overlapping regions.

Algorithm 4 *2LSB-2b exclusion* $[\mathcal{I}_{2b}, \mathcal{E}'_{2a}] = \mathrm{I}_{2b}(\mathcal{E}_{2a}, \mathcal{W}_d)$

$\mathcal{I}_{2b} \leftarrow \emptyset$
$\mathcal{E}'_{2a} \leftarrow \emptyset$
for all $E \in \mathcal{E}_{2a}$ **do**
  for all $W \in \mathcal{W}_d$ **do**
    if $W \cap_p E \neq \emptyset$ **then**
      $\mathcal{E}'_{2a} \leftarrow \mathcal{E}'_{2a} \cup \{E\}$
    else
      $\mathcal{I}_{2b} \leftarrow \mathcal{I}_{2b} \cup \{E\}$
    end if
  end for
end for

3.3.4. Opening 2LSB-2a surfaces generation

The opening 2LSB-2a surfaces are generated using the internal and external 2LSB-2a surfaces obtained from the previous steps and the boundary representations of the opening volumes obtained in the data preprocessing stage. This generation is carried out by Algorithm 5 using a special function named maxA. For each opening B-rep $\mathcal{OP}$, the function maxA selects the polygon $O \in \mathcal{OP}$ with the largest area among those whose unit normal vector $\hat{n}_o$ is parallel or antiparallel to a given direction $\hat{n}$ (i.e., $|\langle \hat{n}_o, \hat{n} \rangle| = 1$). This ensures that the selected polygon is the largest face aligned with the given orientation $\hat{n}$. Mathematically, this is expressed as:

$$O = \mathrm{maxA}(\mathcal{OP}, \hat{n}) \tag{5}$$

As detailed in Algorithm 5, each polygon B in the internal ($\mathcal{I}_{2a}$) and external ($\mathcal{E}_{2a}$) 2LSB-2a sets is paired with the representative opening polygon O obtained via maxA. Polygon O is the opening B-rep polygon with the largest area from the set $\mathcal{OP}_j$ aligned with the normal $\hat{n}_B$ ($O = \mathrm{maxA}(\mathcal{OP}_j, \hat{n}_B)$). The intersection-pair function IP_d is then applied to (B, O) and $(B, -O)$ pairs to ensure correct handling of polygon orientation. Any non-empty intersection polygons returned by these operations are appended to the output set $\mathcal{O}_{2a}$. Thus, the algorithm extracts all opening-related 2LSB-2a surfaces that lie on the boundary B. Fig. 8 illustrates this process visually, showing how B intersects with the O projection via the IP_d function and produces the opening 2LSB-2a surface.

Algorithm 5 *Opening 2LSB-2a generation* $\mathcal{O}_{2a} = \mathrm{O}_{2b}(\mathcal{I}_{2a}, \mathcal{E}_{2a}, \{\mathcal{OP}_1, ..., \mathcal{OP}_M\})$

$\mathcal{O}_{2a} \leftarrow \emptyset$
for $j \in \{1, ..., M\}$ **do**
  for all $B \in \{\mathcal{I}_{2a}, \mathcal{E}_{2a}\}$ **do**
    if $\mathrm{maxA}(\mathcal{OP}_j, \hat{n}_B) \neq \emptyset$ **then**
      $O = \mathrm{maxA}(\mathcal{OP}_j, \hat{n}_B)$
      $[B_{1p}, O_{1p}] = \mathrm{IP}_d(B, O)$
      if $B_{1p} \neq \emptyset$ **then**
        $\mathcal{O}_{2a} \leftarrow \mathcal{O}_{2a} \cup B_{1p}$
      end if
      $[B_{2p}, O_{2p}] = \mathrm{IP}_d(B, -O)$
      if $B_{2p} \neq \emptyset$ **then**
        $\mathcal{O}_{2a} \leftarrow \mathcal{O}_{2a} \cup B_{2p}$
      end if
    end if
  end for
end for

It is important to note that because the internal 2LSB-2a surfaces are processed in pairs, the corresponding internal 2LSB-2a opening surfaces are also generated in pairs. Specifically, the maxA surface $O = \mathrm{maxA}(\mathcal{OP}_j, \hat{n}_B)$ of the internal-opening B-rep is projected by the operations $[B_{1p}, O_{1p}] = \mathrm{IP}_d(B, O)$ and $[B_{2p}, O_{2p}] = \mathrm{IP}_d(B, -O)$, onto both internal 2LSB-2a surfaces $B \in \{\mathcal{I}_{2a}, \mathcal{E}_{2a}\}$ as these two surfaces lie within the threshold distance d.

3.4. Output stage

As illustrated in Fig. 4, the output stage of the proposed methodology encodes the BEM geometric data into various file formats, including XML [36], OBJ [37], and gbXML [38]. In all cases, the simulation zones defined in the output files maintain a one-to-one correspondence with the spatial volumes from the input BIM data.

For the gbXML format specifically, internal 2LSB surface pairs are represented by a single 2LSB surface. This is achieved by computing a middle-plane surface, created by projecting each pair of opposite and parallel 2LSB-2a surfaces onto a plane equidistant from both.

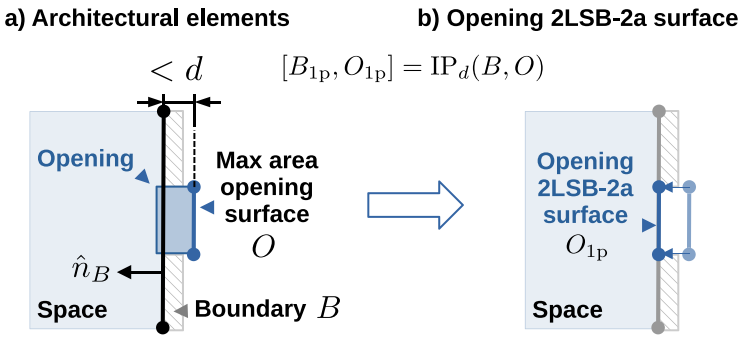


Fig. 8. Opening 2LSB-2a surface generation example from boundary (B), and maximum area opening (O) surfaces.

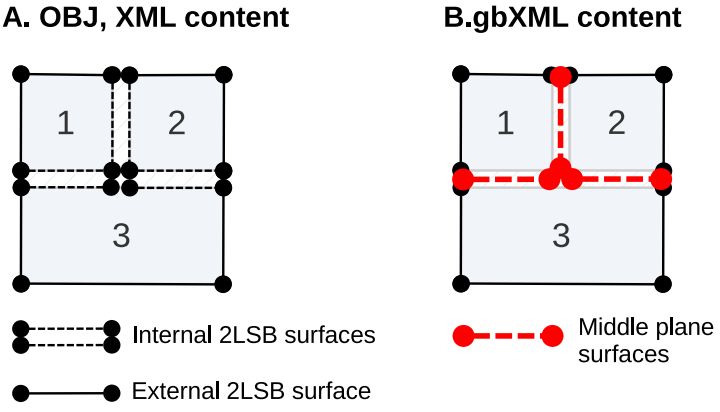


Fig. 9. Example instances of XML, OBJ and gbXML output files.

Fig. 9 illustrates the resulting geometry: part A shows the OBJ and XML outputs, where the original internal 2LSB-2a surface pairs are preserved, while part B depicts the gbXML output, in which each surface pair is replaced by a single middle-plane surface, shown as a red dashed line.

4. Computational complexity and accuracy

The stages of the proposed algorithm have uneven computational costs. For example, detecting internal 2LSB-2a surface pairs requires an outer double loop over all combinations of space volumes, and an inner loop that examines every surface pair in which one surface belongs to the first space and the other to the second. Assuming that the total number of surfaces for space $i \in \{1, \dots, N\}$ is n_i, the number of surface pair checks required to identify the internal 2LSB-2a pairs is

$$T = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} n_i n_j \tag{6}$$

This is because each unordered pair of spaces (i, j) contributes all cross-combinations of their surfaces. This can be rewritten as:

$$T = \frac{1}{2}\left[N_{tot}^2 - \sum_{i}^{N} n_i^2\right] \tag{7}$$

where $N_{tot} = \sum_i^N n_i$ is the total number of space B-rep surfaces. Since $n_i^2 \geq 0$ it is true that:

$$T \leq \frac{N_{tot}^2}{2} \tag{8}$$

Additionally, assuming that the most complex space volume has a maximum number of surfaces $n_{max} = \max_i n_i$, a looser bound of T can be expressed as:

$$T \leq \frac{N(N-1)}{2} n_{max}^2 \tag{9}$$

For external 2LSB-2a surface detection, the computational expense is dominated by CGAL's alpha-wrapping procedure [39]. Each external surface must be compared against each triangle of the alpha wrap to identify true external surfaces and distinguish them from 2LSB-2b surfaces. The total number of alpha-wrap triangles, which drives this computational cost, depends on both α and the offset parameter. No general relationship exists between the total number of space surfaces and the number of alpha-wrap triangles, as this is influenced by the internal complexity of the building topology; buildings with more intricate interiors may have a greater number of space polygons relative to the alpha-wrap triangles.

Lastly, for extracting 2LSB-2a surfaces from opening volumes, the algorithm processes all M openings to identify the maximum area face of each. Each such maximum area face is then checked for intersections of projections on all N_{2LSB} previously identified non-opening 2LSB-2a surfaces. This results in a computational complexity of $\mathcal{O}(M N_{2LSB})$.

Finally, before performing polygon subtraction and intersection/projection operations on a given polygon pair (A,B), two geometric conditions are evaluated: coplanarity and opposing orientation. Specifically, the dot product of the unit normal vectors $\hat{n}_A$ and $\hat{n}_B$ and the distances d_A and d_B from the origin of the polygons' hosting planes, are checked to satisfy $|\langle \hat{n}_A, \hat{n}_B \rangle| = 1 \wedge d_A = d_B$ (coplanarity) and $\langle \hat{n}_A, \hat{n}_B \rangle < 0$ (opposite direction). In practice, however, exact coplanarity and perfectly opposing unit normal vectors are uncommon due to numerical precision limits and modeling inaccuracies. Therefore, these conditions are relaxed using a tolerance parameter $\epsilon > 0$, resulting in the modified checks: $|\langle \hat{n}_A, \hat{n}_B \rangle| > 1 - \epsilon$ for approximate coplanarity and $\langle \hat{n}_A, \hat{n}_B \rangle < -\epsilon$ for approximate opposite direction. To interpret ϵ geometrically note that the coplanarity condition $|\langle \hat{n}_A, \hat{n}_B \rangle| > 1 - \epsilon$ corresponds to an angular deviation $\theta < \arccos(1-\epsilon)$. For instance, $\epsilon = 10^{-3}$ yields $\theta < 2.6°$ and $\epsilon = 10^{-2}$ yields $\theta < 8.1°$. In our workflow $\epsilon = 10^{-3}$ was used as a practical default.

Finally, this coplanarity threshold serves not only as a numerical precision adjustment but also as an inherent requirement of the method, ensuring correct surface classification when distinguishing external 2LSB-2a surfaces from 2LSB-2b surfaces based on the alpha-wrap triangles.

5. Implementation

The proposed workflow is implemented in C++ for efficient handling of large-scale building models and relies on three external libraries, Clipper [40], CGAL [41], RapidXML [42], and components of the BIM management platform (BIM-MP [34]), each supporting specific stages of the sequential processing pipeline, as presented in the following subsections.

5.1. Preprocessing stage dependencies

The preprocessing workflow begins with the Geometry Exporter component of the BIM Management Platform [34], which is responsible for deserializing IFC input models and extracting their geometric content. The exporter produces an XML representation that constitutes a custom serialization of the IFC schema, covering the entities of the IfcGeometryResource, and encodes both procedural and explicit geometry associated with all instances of architectural elements. This XML file is parsed using RapidXML [42], chosen for its low memory footprint and high performance when processing large BIM datasets. The parsed data are mapped into internal C++ data structures representing the architectural elements.

Geometry preprocessing then converts all procedural definitions into explicit boundary representations. This conversion is carried out using constructive solid geometry (CSG) Boolean operations provided by CGAL, specifically its Nef polyhedra kernel [41], which ensures robustness and exactness in solid modeling operations. These operations are carried out by the B-rep Generation (BRG) module of the BIM Management Platform, which performs the transformation from procedural and boundary representations to explicit polygonal B-reps.

Explicit geometries that are already defined as boundary representations are transferred directly without further conversion. The resulting B-reps are guaranteed to be closed, consistently oriented, and suitable as input for the 2LSB surface computation stage described in Section 3.2.

As an alternative to the geometric data preprocessing pipeline, the open-source library IfcOpenShell [43] can be used to perform much of its functionality. In particular, IfcOpenShell provides direct access to IfcSpace and IfcOpeningElement entities and supports the extraction and tessellation of their geometric representations into boundary representations (B-reps). Its geometry engine can therefore be used to obtain closed surface meshes with consistent orientation, fulfilling the geometric prerequisites outlined above.

5.2. Main processing stage dependencies

To compute the set of 2LSB surfaces, the workflow performs extensive polygon intersection and subtraction operations (denoted by $\setminus_p$ and $\cap_p$). These operations are executed in the main part of the workflow. They are implemented using the Clipper library [44], chosen for its numerical robustness, high-performance Boolean operations, and its support for complex polygon structures in which an outer polygon may contain holes, those holes may in turn contain polygons, and so on, forming a hierarchy of nested regions with finite depth. Finally, to exclude 2LSB-2b surfaces, a closed enclosure of all space-volume solids is generated using an alpha-shape-based method, implemented via CGAL's Alpha Wrap routine [39].



Fig. 10. Photo of UCL's OPS building.

6. Geometric data prerequisites

To ensure compatibility with the algorithm, the input geometric data should meet the following key conditions:

1. Spaces and openings should be modeled explicitly, either through procedural definitions or as closed B-rep solids, depending on the export format of the software used to model these elements. IFC exporters meet this requirement by default. Other inputs, are also acceptable as long as they can be transformed into boundary representations with outward-pointing surface normal vectors.
2. The space geometries must be geometrically clean and topologically valid, avoiding issues such as self-intersections, open shells, or inconsistent face orientations. In contrast, opening volumes are not subject to these strict constraints. This can be achieved by using an element clash checker [45] to detect and resolve potential space intersections in the model prior to its use in the methodology. Surface inversions can be identified with a standard element checker.
3. The coordinate system must be appropriately geo-referenced so that all building space and opening volumes are expressed in the same unified spatial context. This is typically achieved by exporting the IFC files from a single, unified BIM authoring tool file.

The accuracy of the proposed algorithm may deteriorate significantly when provided with low-quality or geometrically inconsistent data.

7. Case studies and validation

The proposed workflow was implemented in two buildings: One is a new mixed-use building, named One Pool Street (OPS), at the UCL east campus, London, UK. With a total floor area of over 17,000 m^2, it includes student accommodation and integrated office spaces. The other is an older (1980) 2500 m^2 building at EDF Lab Les Renardières in Moret-Loing-et-Orvanne, Île-de-France, north-central France. The OPS building, being relatively large with data fragmented across multiple BIMs, was chosen to demonstrate the methodology's capability in processing data from multiple IFC BIMs (three in this case). Meanwhile, the EDF building was selected to showcase the methodology's ability to handle opening volumes shared by different space volumes. These buildings are pictured in Figs. 10 and 11 photos, respectively.

The exported building energy model geometries for the two case study buildings are presented in the next subsection.



Fig. 11. Photo of EDF's building.

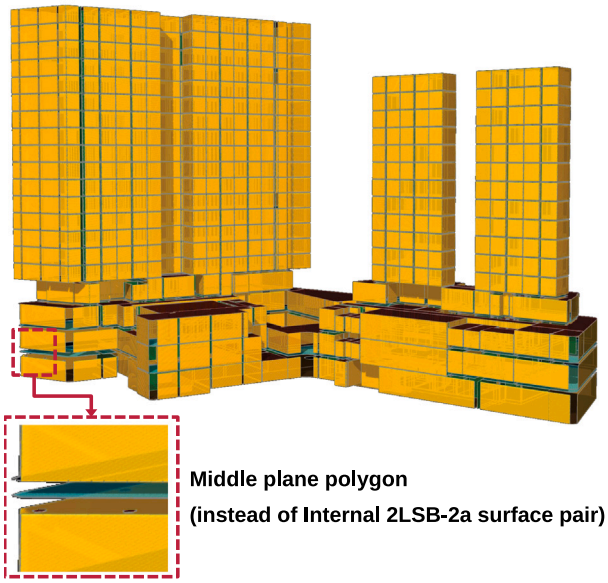


Fig. 12. Visualization of UCL's OPS BEM geometry (gbXML).

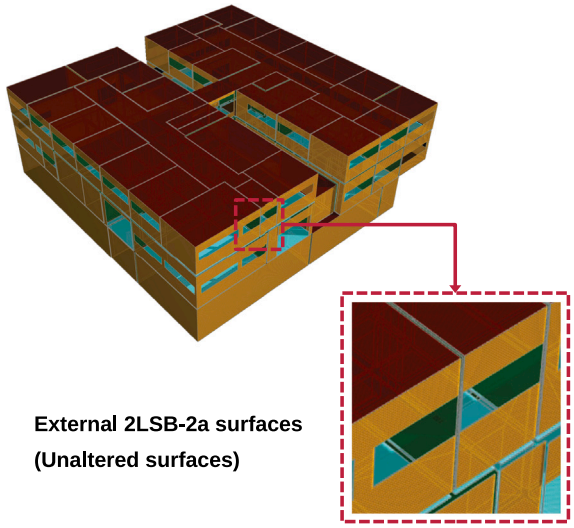


Fig. 13. Visualization of EDF's BEM geometry (gbXML).

Table 1
OPS BEM's 2LSB surface types and numbers.

Internal non-opening surfaces	6200 (3100 pairs)
Internal surfaces	**6200 (3100 pairs)**
External non-opening urfaces	2749
External surfaces	**2749**
Opening surfaces	**0**

7.1. Building energy model geometry export

Applying the proposed methodology to the BIM space and opening geometries of the two case study buildings results in 2LSB-2a surfaces that are subsequently exported into gbXML-based BEM geometries, as shown in Fig. 12 (OPS building) and Fig. 13 (EDF building).

Minor geometric operations are performed to convert the 2LSB-2a surfaces into the BEM geometries for the two case study buildings. These geometric operations are applied after the proposed methodology and involve replacing each pair of internal 2LSB-2a surfaces with a single polygonal surface positioned at the middle plane between the parallel planes of the original surfaces as described in Section 3.4. This replacement is performed algorithmically for gbXML exporting reasons, and is demonstrated in the magnified section of UCL's OPS BEM geometry at the bottom of Fig. 12. Additionally, the external 2LSB-2a surfaces and their associated opening surfaces are transferred during the gbXML export without any additional processing as highlighted in the magnified section of EDF's BEM geometry in Fig. 13. These generated geometries and the parameters used for each case study building are analyzed in subsequent subsections.

7.2. OPS building

The input data for the OPS building consist of three IFC BIM files corresponding to the east tower, west tower, and podium. These files were deserialized using the Geometry Exporter, and all 834 geometric descriptions of the building's space volumes (450 from West Tower, 218 from East Tower and 166 from Podium), were merged into a single XML file during the preprocessing stage of the proposed methodology (see Section 3). These descriptions were then converted into B-reps, which are displayed in different colors in the left part of Fig. 14.

All 834 space B-reps identified in the input XML file were processed using the four-step methodology outlined in the main processing stage, resulting in the generation of a BEM geometry comprising 834 thermal zones, as shown on the right side of Fig. 14. This BEM geometry includes both internal and external 2LSB-2a surfaces, which are differentiated by color in the same figure. The internal 2LSB-2a surfaces are depicted in gray, while the external ones appear in yellow. Due to the OPS building's thick floor slabs, a thickness threshold of $d = 1.5$ m was applied to prevent the alpha wrap surfaces from intruding into the space between the building space volumes. In this case study both alpha and offset parameters were 10^{-3} and the number of produced alpha-wrap triangles were 514,499.

By zooming into specific areas of the 2LSB-2a surface set, it can be observed that the proposed methodology accurately identifies internal and external 2LSB-2a surfaces related to horizontal slab partitions between the tower and the podium (Zoom 1: tower base) and vertical wall partitions attached to misaligned space volumes (Zoom 2).

OPS building was simulated using EnergyPlus [46] without opening volumes since opening volume definitions were missing from the BIM input data. Although the expected simulation results are unrealistic due to the omission of opening-related effects, the simulations were conducted to demonstrate that the proposed approach produces a simulation-ready BEM. Fig. 15 presents the 3D visualization of the EnergyPlus model. Table 1 lists the total surfaces generated for each 2LSB surface type, by the proposed workflow. On a 6th generation i7 processor, the entire operation required 2393 s in total, with 455 s spent on the main operations and 1938 s on the alpha-wrap calculations.

7.3. EDF building

In contrast to OPS, the EDF IFC BIM includes information on opening volumes. To demonstrate the methodology's capability in handling

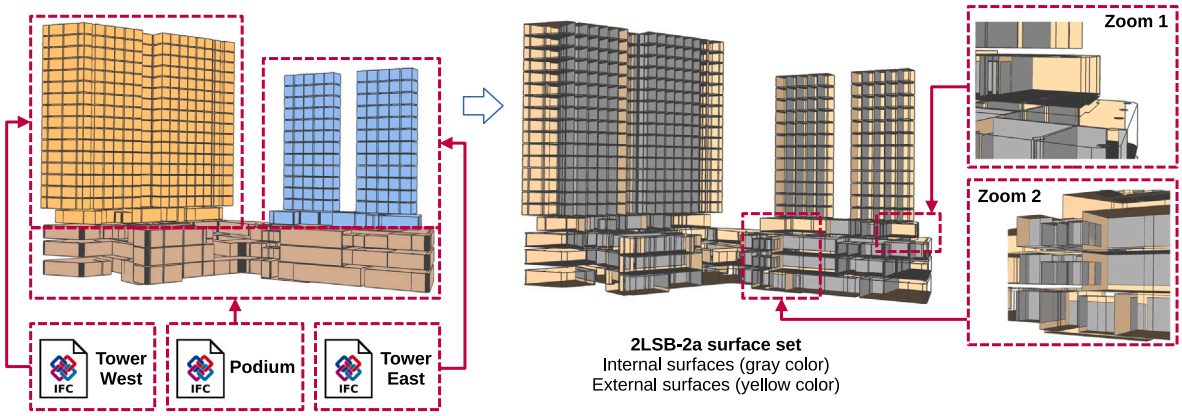


Fig. 14. 2LSB-2a generation process for UCL's OPS building, using space geometry data from three input IFC files.

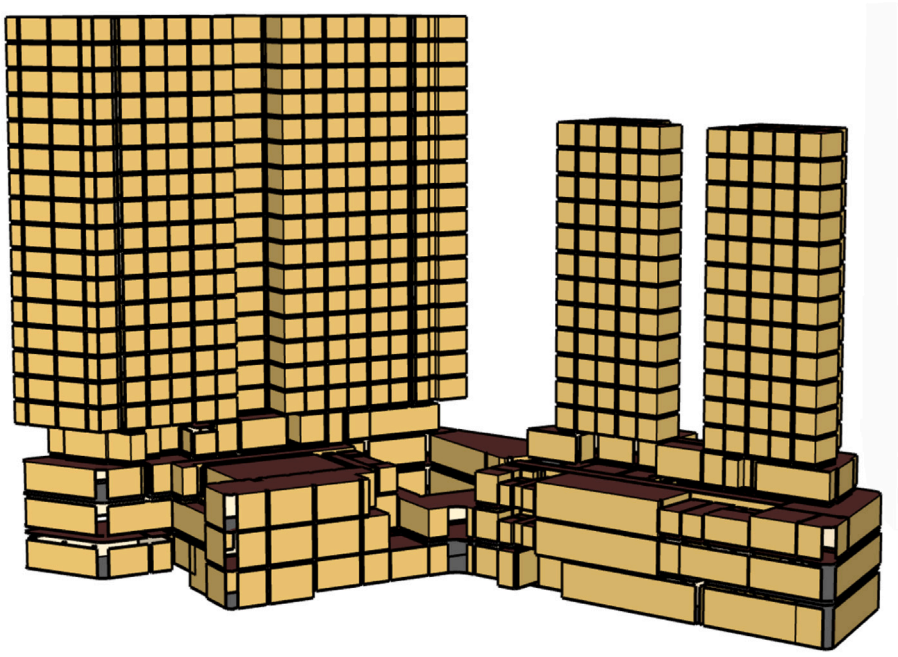


Fig. 15. 3D visualization of the EnergyPlus model for OPS building.

opening volumes for generating the surface set $\mathcal{O}_{2a}$, the EDF's opening B-reps, along with the building's space B-reps (shown in the left part of Fig. 16), were used as input. After applying the methodology a 2LSB-2a surface set is produced. The total number of surfaces for each 2LSB type is reported in Table 2 and displayed in the right panel of Fig. 16. For this building case a thickness threshold of $d = 0.6$ m was set and 80 space volumes in the input BIM data produced a BEM geometry consisting of 80 thermal zones. In this case study, both alpha and offset parameters were 10^{-3} and the number of produced alpha-wrap triangles were 48,897.

Two types of building configurations require further attention, pictured in the Zoom 1, 2, and 3 magnified areas in Fig. 16. In the first type, a single opening is used to generate two opening 2LSB-2a surfaces associated with the adjacent spaces to which the opening belongs. Although the opening itself is singular, and the maximum area surface (O) returned by the function maxA is unique, it is projected onto two surfaces (B) from the adjacent space B-reps. These projections are then intersected with the surfaces of these spaces using the function $\mathrm{IP}_d(B, O)$, resulting in two adjacent opening 2LSB-2a surfaces, as shown in the Zoom 1 and 2 areas, in the right part of Fig. 16. The second building configuration type, demonstrated in the Zoom 3 area in Fig. 16, illustrates that the proposed methodology correctly generates the internal 2LSB-2a opening surfaces corresponding to internal doors.

Table 2
EDF BEM's 2LSB surface types and numbers.

Internal non-opening surfaces	696 (348 pairs)
Internal surfaces	**886 (443 pairs)**
External non-opening surfaces	221
External surfaces	62
Opening surfaces	**349**

The total number of generated internal, external, opening and non-opening 2LSB surfaces are summarized in Table 2. The total execution time on a 6th generation i7 processor was 43 s, of which 42 s were dedicated to the alpha-wrap calculations.

8. Discussion

The previously presented simulation results demonstrate that the proposed method effectively and accurately generates the geometric content required for building energy models (BEMs), ensuring that

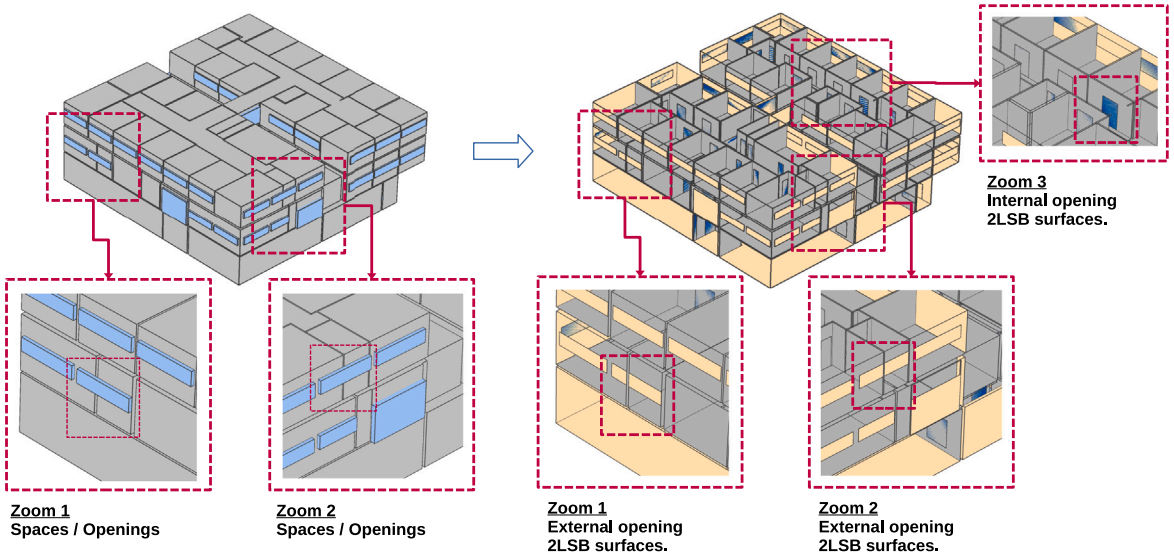


Fig. 16. 2LSB-2a generation process for EDF's building, using space and opening geometry data.

the resulting data can be seamlessly integrated into various energy simulation workflows. This capability is crucial for conducting detailed performance analyses of buildings, enabling designers, engineers, and researchers to assess energy efficiency across different scenarios. By leveraging the proposed workflow, users can streamline the process of BEM creation, reducing manual efforts and minimizing the risk of errors that may arise from model inconsistencies, data translation issues and design errors related to building constructions.

In this paper, the generated BEM geometric content was exported in gbXML format, which served as an intermediate data interface to facilitate interoperability between modeling and simulation environments. The adoption of gbXML is particularly beneficial, as it is widely supported by various building information modeling (BIM) and energy simulation tools, allowing for efficient data exchange and eliminating the need for labor-intensive model reconstruction. This capability enhances workflow automation and supports more effective decision-making in building performance analysis.

Furthermore, the generated geometric content is not limited to gbXML exportation but can also be converted into other widely used formats to accommodate specific simulation requirements. For example, the content can be transformed into IDF, the native input file format for EnergyPlus, enabling direct use within this simulation environment. This conversion process relies on appropriate transformation tools, such as the one developed in [34], which ensures that the necessary geometric and semantic information is accurately mapped from one format to another. The feasibility of this conversion is contingent on utilizing the workflow's native XML output format as an input, ensuring consistency in data exchange and maintaining the integrity of the simulation-ready model. By supporting multiple output formats, the proposed method enhances flexibility, making it a valuable asset for energy modelers working across different software ecosystems.

Finally, a significant portion of the total computation time is devoted to the alpha wrap calculations of the space volume geometries. For the OPS building, these calculations accounted for approximately 81% of the total time (1938 out of 2393 s), while for the EDF building they represented about 97% (42 out of 43 s). This is because the alpha wrap computation relies on an external CGAL routine, which is considerably more computationally demanding than the projection and intersection operations that are internally optimized and implemented in C++. To further reduce the alpha wrap computation time, alternative approaches could be explored in future work.

9. Limitations and future directions

Although the proposed methodology successfully generates the BEM geometry for most of the IFC files, it has limitations concerning errors in space geometries, curved geometric external shading surfaces, construction material data, and very thin adjacent space volumes, as discussed in the following subsections.

9.1. Incorrect space geometries

The geometric accuracy of IfcSpace entities can vary considerably across BIM authoring tools. For example, spaces may not extend fully to the ceiling, may be misaligned beneath sloped roof surfaces, or may contain unintended voids. To address these issues, the proposed method requires that the input space volumes be non-overlapping and topologically valid. Prior to processing, geometric consistency checks that include space volume clash detection, space shell-validity verification, and space visual inspection, may be applied to ensure that each space volumes form closely located enclosures. Enforcing these prerequisites significantly reduces the risks associated with imperfect space definitions and helps ensure that the derived 2LSB surfaces remain robust and reliable.

9.2. Curved and non-convex geometries

Surfaces in the output 2LSB-2a set with relatively small areas contribute minimally to the building energy model, while increasing simulation time with negligible impact on accuracy [27]. These small surfaces typically arise as minor recesses in highly non-convex regions or along curved boundaries, where curvature is approximated by several adjacent polygonal faces. To address this, surfaces whose areas fall below a minimum threshold are removed. Applying an area threshold inevitably creates small gaps in the 2LSB-2a surface set at curved boundary locations. These omissions, clearly visible at the rounded corners of UCL's OPS building, are illustrated in the left part of Fig. 17. In the proposed workflow, a minimum area threshold of $A_{\min} = 0.25\ \mathrm{m}^2$ is used, that corresponds approximately to a $50,\mathrm{cm} \times 50,\mathrm{cm}$ patch, which is well below the spatial resolution typically relevant for thermal exchange in building energy models. When curved regions or deeply indented non-convex boundaries are approximated using many planar faces, the dihedral angles along the common edges of the B-rep mesh become small, (this is illustrated in the right part of Fig. 17, where smaller dihedral angles are related to finer approximations). This

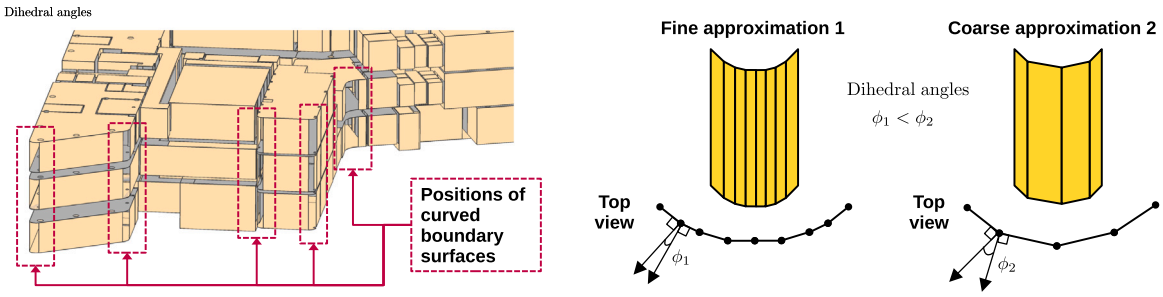


Fig. 17. Positions of curved boundary surfaces and respective 2LSB-2a surface exclusion in the podium of UCL's OPS building (left) and fine/coarse approximations and their dihedral angles (right).

results in a large number of small adjacent surface patches. If the curved or non-convex region is extensive (as in circular or highly curved façades), applying a simple area-threshold criterion may remove all of these patches, unintentionally eliminating the entire region from the 2LSB-2a surface set. This behavior represents a significant limitation of the current method.

To address this limitation, in future versions of the workflow, a B-rep simplification step, based on edge-collapse operations constrained by a minimum dihedral-angle threshold, will be introduced. This approach ensures that curved and non-convex areas are represented using a coarser mesh before area filtering is applied. As a result, these regions are preserved through coarser, larger surface elements, avoiding the unintended elimination that occurs when the mesh is discretized too finely, and preserving the geometric continuity of the 2LSB-2a surface set, at the location of the curved regions.

9.3. External shading surfaces

The external shading surfaces of a building affect energy simulation by reducing thermal gains from direct solar irradiance. However, since the proposed workflow is based on building spaces rather than constructions, it cannot identify these shading surfaces solely from the space volumes. In case there are external shading surfaces, building constructions should be included in the calculations. Such investigation is a subject of future work.

9.4. Missing opening volumes

Some IFC files, such as the one used in the first case study, do not include explicit opening volume definitions. In these situations, the workflow cannot produce opening-related 2LSB surfaces. A potential extension of the proposed methodology could involve synthesizing opening geometries from construction elements, such as windows, doors, or curtain wall panels. However, this functionality is not currently implemented, as reconstructing openings from these elements introduces additional challenges, including assumptions about parent-wall thickness, surface alignment tolerances, and potential inconsistencies in opening element placements. These factors would add considerable geometric complexity and fall outside the present scope of the paper.

9.5. Curtain walls

Curtain walls are a specific type of building construction that may include transparent surfaces, allowing solar gains to enter the internal space volumes. To integrate curtain walls into the proposed workflow, special handling is necessary. Only the transparent plate elements should be considered as openings and included in the workflow's calculations, while the opaque elements should be excluded. Future versions of the workflow could consider converting the transparent plate elements of curtain walls into opening volumes. Through this conversion, curtain walls could be treated as openings and thereby integrated into the proposed workflow. Since the impact of curtain wall member elements on the simulation results is negligible, these elements can be omitted.

9.6. Adjacent thin space volumes

In rare cases where two adjacent building space volumes are extremely thin—having a combined thickness significantly below the threshold defined by the proposed methodology (see floor plan in part (a) of Fig. 18)—the algorithm may erroneously generate internal 2LSB-2a surface pairs by projecting and intersecting, via the IPS_d function, two polygons $A \in SP_1$ and $B \in SP_2$, each belonging to adjacent spaces 1 and 2 and having opposite unit normal vectors, as illustrated in part (b) of Fig. 18. Since such cases are extremely rare and cannot be eliminated, users of the methodology are encouraged to manually detect and remove such incorrectly generated 2LSB-2a surface pairs using a graphical user interface (GUI) and the visual OBJ file output of the methodology.

9.7. Construction materials

To perform a building energy simulation without active building systems, the thermal properties of the materials, such as thermal transmittance, for the building constructions are necessary. These properties are associated with the 2LSB-2a surfaces and correspond to the construction materials of the building elements to which the 2LSB-2a surfaces are attached. In traditional 2LSB surface generation algorithms, these properties are extracted from the IFC BIM data. However, in this case, since the input data of the workflow does not include any building constructions, these properties must be defined manually [2]. Four types of properties need to be manually specified, covering all combinations of internal/external and opening/non-opening constructions.

10. Conclusions

This paper presented an approach for generating the second-level space boundary surface set of Building Energy Models (BEM). The proposed method addresses key limitations of traditional methods that primarily rely on the geometry of architectural construction elements such as walls and slabs. By shifting the focus to space and opening volumes instead, our methodology reduces dependence on the complexity of building constructions and minimizes the impact of human-induced design errors. This approach streamlines the building energy modeling process by requiring only a predefined construction thickness threshold and intersection-free space volumes, making it both computationally efficient and practical for integration into automated workflows.

The proposed method was validated in two case studies, a university building and an office building, demonstrating its capability to handle complex geometries, including intricate spatial configurations and diverse opening volume topologies. The results confirm that our

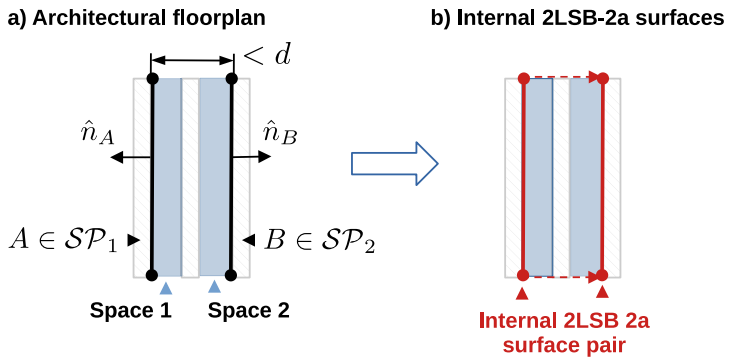


Fig. 18. Incorrect internal 2LSB 2a surfaces detected on adjacent very space volumes.

approach enhances the accuracy of BEM geometric generation and significantly improves automation in the modeling process. By leveraging federated BIM input data, the method exhibits scalability, ensuring its applicability across a wide range of large and complex building designs.

At the same time, it is important to acknowledge that the method is not fully self-contained. Its practical application relies on two complementary components: (a) an efficient method for determining exterior polygons, which remains a computationally challenging problem, and (b) a designer or modeler preparing clean input data, deriving appropriate space and opening volumes from BIMs that are typically created from architectural elements. While our approach robustly solves the geometric processing step given these prerequisites, its performance and applicability depend on the quality and preparation of input data and the availability of suitable exterior detection techniques.

Beyond its immediate contributions to geometric data generation in energy modeling, our work has broader implications for the future of automated and intelligent BEM workflows. The ability to generate precise and robust second-level space boundaries with minimal manual intervention paves the way for more efficient energy simulations, supporting better-informed decision-making in sustainable building design. Furthermore, improved geometric accuracy can enhance interoperability with widely used simulation engines, such as EnergyPlus and Modelica, fostering more seamless integration between BIM-based design processes and building energy performance analysis tools.

Future research directions could explore additional refinements to enhance the adaptability of our method across diverse building typologies, including irregular or highly complex architectural forms. Moreover, integrating this approach with existing BEM workflows and expanding support for various BIM standards and data exchange formats could further improve its interoperability and usability in real-world applications. Another promising avenue involves leveraging machine learning or rule-based automation to optimize construction thickness parameter selection, reducing manual input requirements. By addressing these aspects and the dependencies on exterior polygon detection and clean input preparation, our approach could contribute to a more intelligent, scalable, and widely applicable framework for automated BEM generation, ultimately advancing the field of building energy modeling and simulation.

CRediT authorship contribution statement

Georgios Nektarios Lilis: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mathias Bouquerel:** Resources, Writing – original draft, Writing – review & editing, Validation. **Meng Wang:** Writing – review & editing, Writing – original draft, Validation. **Kyriakos Katsigarakis:** Writing – review & editing, Writing – original draft, Resources, Data curation. **Ivan Korolija:** Writing – review & editing, Writing – original draft. **Dimitrios Rovas:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT-4 in order to improve grammar, correct errors, and enhance readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Horizon Europe DigiBUILD project under contract No. 101069658, and co-funded by the UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding guarantee grant number 10040988. The authors would like also to acknowledge EDF's team members Hasan Sayegh and Thierry Duforestel for their contributions to the modeling work related to EDF's building.

Data availability

Data will be made available on request.

References

[1] V. Bazjanac, Space boundary requirements for modeling of building geometry for energy and other performance simulation, in: Proceedings of the CIB W78: 27th International Conference, Cairo, Egypt, 2010, no DOI available. URL https://itc.scix.net/paper/w78-2010-127.

[2] H. Kim, Z. Shen, I. Kim, K. Kim, A. Stumpf, J. Yu, BIM IFC information mapping to building energy analysis (BEA) model with manually extended material information, Autom. Constr. 68 (2016) 183–193, http://dx.doi.org/10.1016/j.autcon.2016.04.002.

[3] X. Li, J. Wen, Review of building energy modeling for control and operation, Renew. Sustain. Energy Rev. 37 (2014) 517–537, http://dx.doi.org/10.1016/j.rser.2014.05.056.

[4] K.E.A. Ohlsson, T. Olofsson, Benchmarking the practice of validation and uncertainty analysis of building energy models, Renew. Sustain. Energy Rev. 142 (2021) 110842, http://dx.doi.org/10.1016/j.rser.2021.110842.

[5] U.S. Department of Energy, Tips and tricks for using EnergyPlus, 2022, EnergyPlus Version 22.1.0 documentation. URL https://energyplus.net/assets/nrel_custom/pdfs/pdfs_v22.1.0/TipsAndTricksUsingEnergyPlus.pdf.

[6] M. Regidor, M. Andrés, S. Álvarez, BIM-to-BEPS conversion tool for automatic generation of building energy models, in: eWork and eBusiness in Architecture, Engineering and Construction, CRC Press, 2018, pp. 73–78, http://dx.doi.org/10.1201/9780429506215-10.

[7] H. Sayegh, G.N. Lilis, M. Bouquerel, T. Duforestel, K. Katsigarakis, D. Rovas, Automatic modelica BEM generation from IFC BIM, in: 2024 IEEE International Workshop on Metrology for Living Environment, MetroLivEnv, IEEE, 2024, pp. 437–441, http://dx.doi.org/10.1109/MetroLivEnv60384.2024.10615614.
[8] H. Gao, C. Koch, Y. Wu, Building information modelling based building energy modelling: A review, Appl. Energy 238 (2019) 320–343, http://dx.doi.org/10.1016/j.apenergy.2019.01.032.
[9] A. Farzaneh, D. Monfet, D. Forgues, Review of using Building Information Modeling for building energy modeling during the design process, J. Build. Eng. 23 (2019) 127–135, http://dx.doi.org/10.1016/j.jobe.2019.01.029.
[10] R.J. Hitchcock, J. Wong, Transforming IFC architectural view BIMs for energy simulation, in: Proceedings of Building Simulation 2011: 12th Conference of IBPSA, IBPSA, 2011, pp. 1089–1095, http://dx.doi.org/10.26868/25222708.2011.1391.
[11] H. Li, J. Zhang, S. Chang, A. Sparkling, BIM-based object mapping using invariant signatures of AEC objects, Autom. Constr. 145 (2023) 104616, http://dx.doi.org/10.1016/j.autcon.2022.104616.
[12] G. Bastos Porsani, K. Del Valle de Lersundi, A. Sánchez-Ostiz Gutiérrez, C. Fernández Bandera, Interoperability between building information modelling (BIM) and building energy model (BEM), Appl. Sci. 11 (5) (2021) 2167, http://dx.doi.org/10.3390/app11052167.
[13] M.H. Elnabawi, Building information modeling-based building energy modeling: investigation of interoperability and simulation results, Front. Built Environ. 6 (2020) 573971, http://dx.doi.org/10.3389/fbuil.2020.573971.
[14] H. Li, J. Zhang, Interoperability between BIM and BEM using IFC, in: Computing in Civil Engineering 2021, 2021, pp. 630–637, http://dx.doi.org/10.1061/9780784483893.
[15] M. Guo, P. Xu, H. Wang, J. Gu, Z. Chen, Building energy modelling based on building information modelling: the remaining problems and a more robust method, in: Proceedings of Building Simulation Conference, IBPSA, 2023, pp. 1022–1030, http://dx.doi.org/10.26868/25222708.2023.1412.
[16] P. Iliadis, E. Bellos, R. Rotas, A. Kitsopoulou, N. Ziozas, N. Nikolopoulos, E. Kosmatopoulos, Comprehensive framework for dynamic energy assessment of building systems using IFC graphs and Modelica, J. Build. Perform. Simul. (2025) 1–22, http://dx.doi.org/10.1080/19401493.2024.2449375.
[17] D. Jansen, V. Richter, L. Maier, J. Frisch, C. van Treeck, D. Müller, Open-source framework for automated generation of building energy performance simulation models and beyond from BIM Data, Autom. Constr. 179 (2025) 106427, http://dx.doi.org/10.1016/j.autcon.2025.106427.
[18] M. Afzal, K. Widding, E. Hjelseth, M. Hamdy, Systematic investigation of interoperability issues between BIM and BEM, in: ECPPM 2022-eWork and eBusiness in Architecture, Engineering and Construction 2022, CRC Press, 2023, pp. 713–720, http://dx.doi.org/10.1201/9781003354222-90.
[19] A. Ciccozzi, A. Santavicca, T. de Rubeis, A. dario, BIM-BEM interoperability for energy analysis: A comparative study of different strategies, Energy Rep. (2025) 4705–4718, http://dx.doi.org/10.1016/j.egyr.2025.04.027.
[20] C. van Treeck, E. Rank, Dimensional reduction of 3D building models using graph theory and its application in building energy simulation, Eng. Comput. 23 (2007) 109–122, http://dx.doi.org/10.1007/s00366-006-0053-7.
[21] C.M. Rose, V. Bazjanac, An algorithm to generate space boundaries for building energy simulation, Eng. Comput. 31 (2) (2015) 271–280, http://dx.doi.org/10.1007/s00366-013-0347-5.
[22] A. Mediavilla, P. Elguezabal, N. Lasarte, Graph-Based methodology for Multi-Scale generation of energy analysis models from IFC, Energy Build. 282 (2023) 112795, http://dx.doi.org/10.1016/j.enbuild.2023.112795.
[23] G.N. Lilis, G.I. Giannakis, D. Rovas, Automatic generation of second-level space boundary topology from IFC geometry inputs, Autom. Constr. (2017) 108–124, http://dx.doi.org/10.1016/j.autcon.2016.08.044.
[24] D. Ladenhauf, K. Battisti, R. Berndt, E. Eggeling, D.W. Fellner, M. Gratzl-Michlmair, T. Ullrich, Computational geometry in the context of building information modeling, Energy Build. 115 (2016) 78–84, http://dx.doi.org/10.1016/j.enbuild.2015.02.056.
[25] T. El-Diraby, T. Krijnen, M. Papagelis, BIM-based collaborative design and socio-technical analytics of green buildings, Autom. Constr. 82 (2017) 59–74, http://dx.doi.org/10.1016/j.autcon.2017.06.004.
[26] E. Fichter, V. Richter, J. Frisch, C. van Treeck, Automatic generation of second level space boundary geometry from IFC models, in: Building Simulation Conference of IBPSA, Bruges, Belgium, 2021, http://dx.doi.org/10.26868/25222708.2021.30156.
[27] H. Ying, S. Lee, Generating second-level space boundaries from large-scale IFC-compliant building information models using multiple geometry representations, Autom. Constr. 126 (2021) 103659, http://dx.doi.org/10.1016/j.autcon.2021.103659.
[28] H. Ying, S. Lee, A rule-based system to automatically validate IFC second-level space boundaries for building energy analysis, Autom. Constr. 127 (2021) 103724, http://dx.doi.org/10.1016/j.autcon.2021.103724.
[29] J.Z. Vetter, W. Huhnt, Transforming building components into a space partition to identify indoor and outdoor spaces in digital building models, in: International Conference on Computing in Civil and Building Engineering, Springer, 2022, pp. 403–413, http://dx.doi.org/10.1007/978-3-031-32515-1_28.
[30] G.N. Lilis, K. Katsigarakis, D. Rovas, Automatic IFC data enrichment with space geometries for Building Energy Performance Simulations, in: Building Simulation 2021, Vol. 17, IBPSA, 2021, pp. 2133–2140, http://dx.doi.org/10.26868/25222708.2021.30888.
[31] G.N. Lilis, K. Katsigarakis, D. Rovas, High-order second-level space boundary surface calculation for building energy performance simulation models, in: ECPPM 2022-eWork and eBusiness in Architecture, Engineering and Construction 2022, CRC Press, 2023, pp. 295–302, http://dx.doi.org/10.1201/9781003354222.
[32] G.N. Lilis, G. Giannakis, K. Katsigarakis, D. Rovas, District-aware Building Energy Performance simulation model generation from GIS and BIM data, in: Proceedings of the 4th Building Simulation and Optimization Conference, IBPSA-England, Cambridge, UK, 2018, pp. 177–184, Available at: https://discovery.ucl.ac.uk/id/eprint/10056593/.
[33] ISO, Industry Foundation Classes (IFC) for Data Sharing in the Construction and Facility Management Industries, Tech. Rep. ISO 16739-1:2024, International Organization for Standardization, 2024, URL https://www.iso.org/standard/84123.html.
[34] K. Katsigarakis, G.N. Lilis, D. Rovas, A cloud IFC-based BIM platform for building energy performance simulation, in: EC3 Conference 2021, Vol. 2, European Council on Computing in Construction, 2021, pp. 350–357, http://dx.doi.org/10.35490/EC3.2021.177.
[35] C. Portaneri, M. Rouxel-Labbé, M. Hemmer, D. Cohen-Steiner, P. Alliez, Alpha wrapping with an offset, ACM Trans. Graph. 41 (4) (2022) 1–22, http://dx.doi.org/10.1145/3528223.3530152.
[36] W3C, XML file format specification, 2025, https://www.w3.org/TR/REC-xml/.
[37] Wavefront_Technologies, OBJ file format specification, 2025, https://www.fileformat.info/format/wavefrontobj/egff.htm.
[38] gbXML organization, Green building XML schema (gbXML), 2025, https://www.gbxml.org/Schema_Current_GreenBuildingXML_gbXML.
[39] P. Alliez, D. Cohen-Steiner, M. Hemmer, C. Portaneri, M. Rouxel-Labbé, 3D alpha wrapping, in: CGAL User and Reference Manual, sixth ed., CGAL Editorial Board, 2024, URL https://doc.cgal.org/6.0.1/Manual/packages.html#PkgAlphaWrap3.
[40] B. Vatti, A generic solution to polygon clipping, Commun. ACM 35 (7) (1992) 56–63, http://dx.doi.org/10.1145/129902.129906.
[41] P. Hachenberger, L. Kettner, 3D boolean operations on Nef polyhedra, in: CGAL User and Reference Manual, sixth ed., CGAL Editorial Board, 2024, URL https://doc.cgal.org/6.0.1/Manual/packages.html#PkgNef3.
[42] M. Kalicinski, RAPIDXML manual, 2025, https://rapidxml.sourceforge.net/manual.html.
[43] IfcOpenShell contributors, IfcOpenShell: An open source IFC toolkit, 2025, https://ifcopenshell.org.
[44] A. Johnson, Clipper2 - Polygon clipping and offsetting library, 2025, https://www.angusj.com/clipper2/Docs/Overview.htm.
[45] G.N. Lilis, M. Wang, K. Katsigarakis, D. Mavrokapnidis, I. Korolija, D. Rovas, BIM-based semantic enrichment and knowledge graph generation via geometric relation checking, Autom. Constr. 173 (2025) 106081, http://dx.doi.org/10.1016/j.autcon.2025.106081.
[46] U.S. Department of Energy and NREL, EnergyPlus, 2025, Available online: https://energyplus.net/.